



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI INHA UNIVERSITY IN TASHKENT

Module Name: Creative Engineering Design
Module Code: SOC1020
Project Type: Group Project (4-6 students)
Project Assigned Date: 08.05.2023
Registered Teams: [View Here](#)
Project Submission Deadline: 31.05.2023 at 23:55
Experiment Dates: 1-2.06.2022
Weighting: 40%

Instructor: Dr. Sarvar Abdullaev
Email: s.abdullaev@inha.uz
Office Hours: Monday, 13:00 - 14:00
Friday, 13:00 - 14:00
Office: Room B411

Learning Outcomes

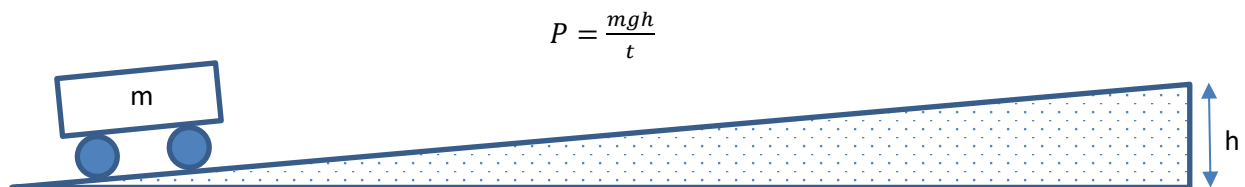
1. Understanding key steps in product design and development process
2. Modelling conceptual design of products and selecting conceptual models for further development
3. Prototyping and testing products according to their detailed specifications
4. Building comprehensive prototypes using 3D printing technologies
5. Designing and conducting experiments with comprehensive prototypes
6. Analysing and modelling the results of experiments

Things to Submit:

1. **PHYSICAL PROTOTYPE** of an electric vehicle as specified in this assignment.
2. **PRINTED and BOUND PRESENTATION SLIDES** as specified in this assignment
3. **PRINTED and SIGNED MEETING MINUTES** for each meeting conducted along with the supplementary materials used or created during the meeting (e.g. literature review, experimental results, used formulas in physics/mathematics, TRIZ resolutions, handwritten sketches, Autodesk Fusion 360 files (.f3d files) or 3D printing format (.stl files) , or any other relevant findings).
4. **ARCHIVE FILE** containing the electronic version of your presentation slides and all supplementary materials (e.g. literature review, experimental results, used formulas in physics/mathematics, TRIZ resolutions, handwritten sketches, Autodesk Fusion 360 files (.f3d files) or 3D printing format (.stl files) , or any other relevant findings) **submitted to eClass.**

Project Overview

You are required to design and build a comprehensive prototype of an electric vehicle using given DC motor and your own 3D printed parts or handcrafted materials. The **main objective of the project** is to maximize the output power (P measured in Watts) of a vehicle while doing a useful work (A measured Joules) through carrying a load (m measured in kilograms) up on a ramp to a certain height (h measured in meters) over a certain time period (t measured in seconds). The **key requirement** of the project is that the only source of mechanical energy should be the battery powered DC motor provided to every team. Teams cannot use other sources of power in their design.



In this project, you will concentrate your entire effort to searching for the right solution. The detailed target product specifications will be provided beforehand and you don't need to compile them yourself. Therefore there is no need for **user research** or **product specification** phases to be accomplished. Also the oral presentation of your project is not required, however you must submit presentation files as specified in this document. Instead, you will present the working prototype of your vehicle for the **FINAL EXPERIMENT** which will evaluate the output power along with other technical requirements given below. Your mark for this project will be calculated based on the quality of the research you have accomplished (demonstrated in your presentation slides) and the output power of your prototype (measured in final experiment).

Vehicle Types

There are various types of vehicles based on their motion principle. Here are some of the most common types:



Figure 1 Wheeled Vehicle

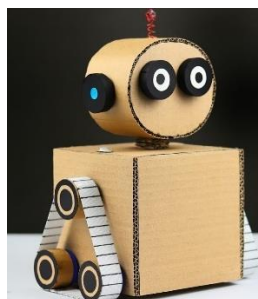


Figure 2 Tracked Vehicle



Figure 3 Legged Vehicle

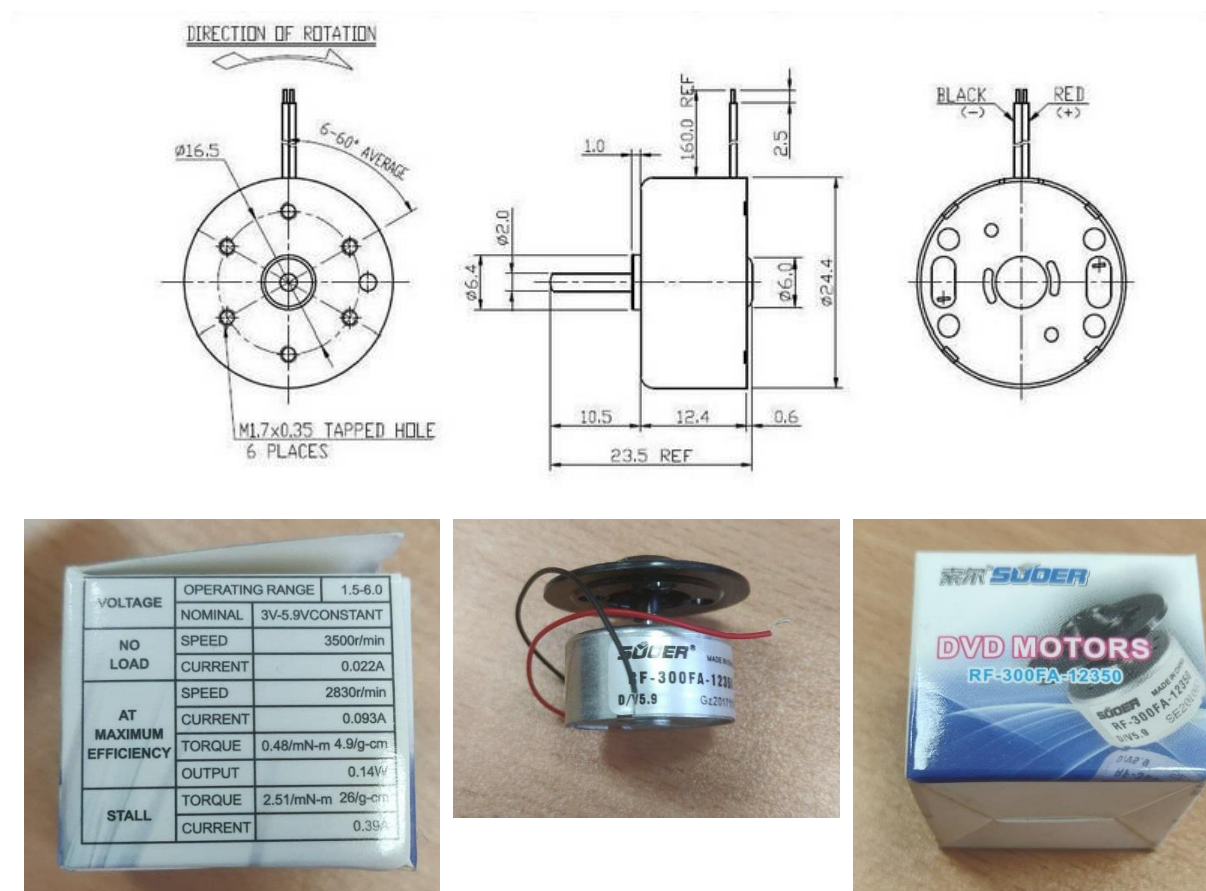
- **Wheeled vehicles:** These are vehicles that move using wheels. They can be two-wheeled or four-wheeled, and they can be powered by different means such as a motor or human power. Examples include cars, motorcycles, bicycles, and scooters. Some DIY examples from Youtube:
 - <https://www.youtube.com/watch?v=w0cjLk41Qqs>
 - <https://www.youtube.com/watch?v=URfpuWqCv5k>
 - <https://www.youtube.com/watch?v=XMg4U5VSlsl>
 - <https://www.youtube.com/watch?v=qymG8w4ZlpE>
- **Tracked vehicles:** These are vehicles that move using tracks rather than wheels. Tracks provide more stability and traction on rough terrain. Examples include tanks, bulldozers, and excavators.
 - <https://www.youtube.com/watch?v=QcBeDbs8n9k>
 - <https://www.youtube.com/watch?v=ga0WCCtxmp8>

- **Legged vehicles:** These are vehicles that move using legs, similar to animals. They are often used in rough terrain where wheels or tracks cannot operate effectively. Examples include quadrupeds, hexapods, and bipeds.
 - <https://www.youtube.com/watch?v=3HTDuWJHADY>
 - <https://www.youtube.com/watch?v=xhghCTpM9QA>
 - https://www.youtube.com/watch?v=2QmOy_HObpo

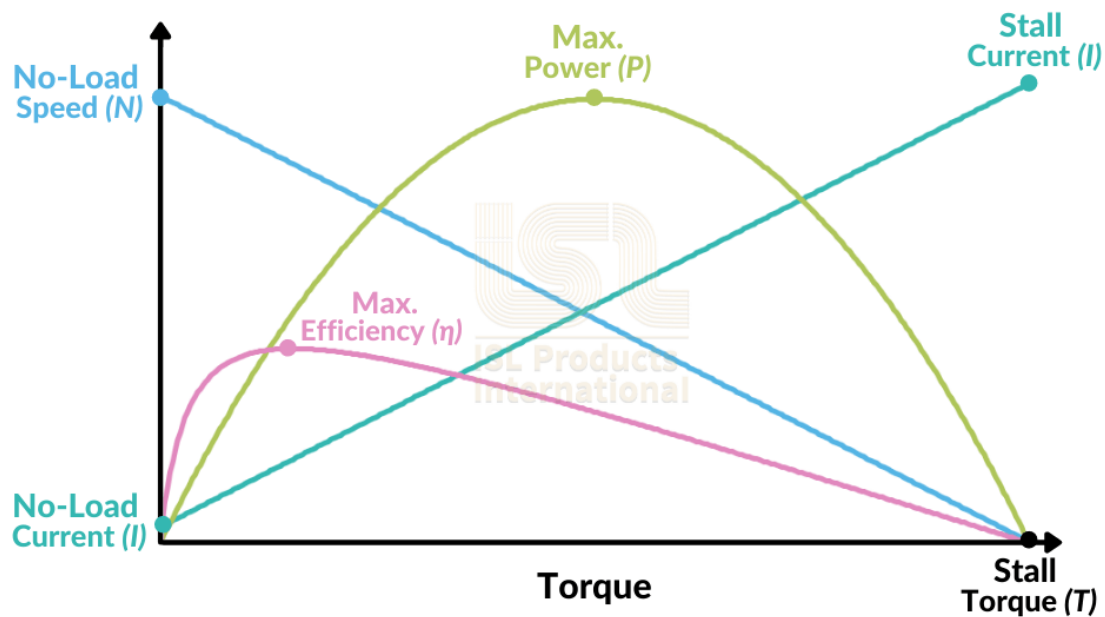
Note that you are not limited to these types of vehicles, and you are free to propose any other type as long as it conforms with the technical requirements given below.

DC Motor

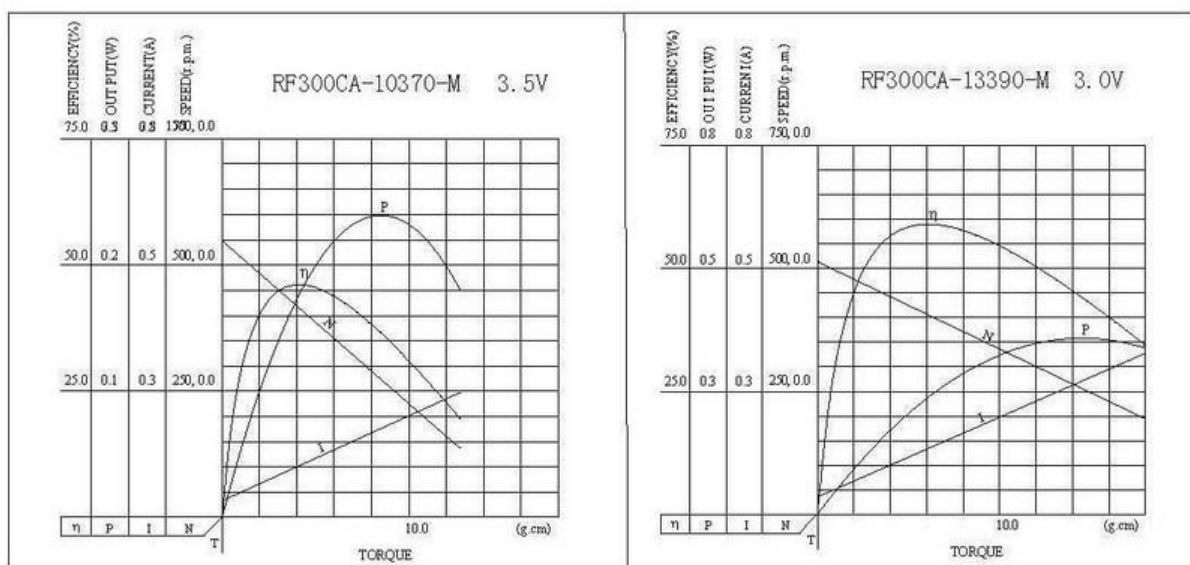
You will be provided with a DC motor with technical specifications shown below. You should use this motor carefully, because no other replacement will be provided in case if it gets damaged. You can find the exact 3D model of the motor enclosed as **motor3.f3d** file to this assignment. **You are not allowed to use any other motor for this project. Using other motors will annul your prototype all together.**



Motor produces around 2200 – 4000 rotations per minute, at maximum efficiency torque is 4.9 g-cms and output 0.14 W. The overall relationship between speed, torque, power and efficiency is given in graph below.



Blue line in graph above shows how speed is related to torque. The greater the amount of load need to be pulled by the motor, the slower it rotates. The moment at which it stops rotating is called Stall Torque. The maximum power (i.e. ability to do work per time unit) is reached when motor can pull sufficiently heavy load at a sufficiently high speed. Green line shows the point when the maximum power is achieved. Diagram below shows this relationship at different voltages for your motor.

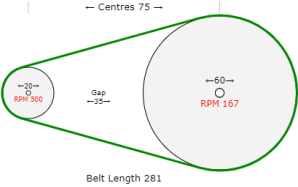
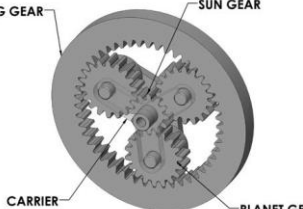


Gears and Belts

The general purpose of gears and belts is to transmit motion and/or power from one shaft to another. In addition, gears and belts have 3 primary usages in a system:

- **Speed ratio:** using different sized gears the input to output rotational speed of shafts can be changed efficiently.
- **Torque ratio:** adjusting the torque ratio of the system.
- **Changing direction:** changing the direction of rotation or converting a rotational motion to a translation motion.

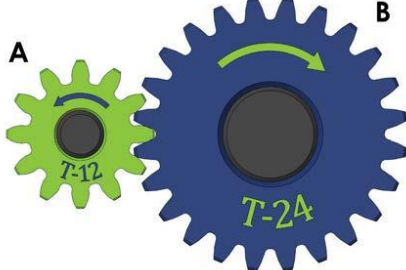
There are several types of gear and belt mechanisms the properties of which described below.

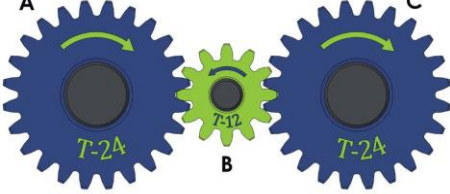
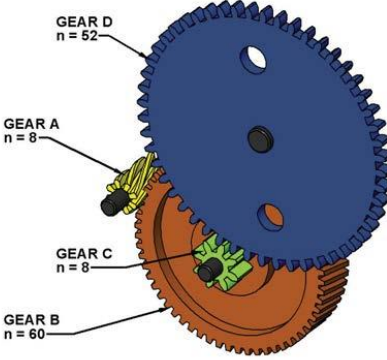
	Belts Belt transmission is a type of power transmission system that uses belts made of materials such as rubber or synthetic polymers to transmit power between two or more pulleys. Although they can slip if the tension is not maintained properly, they could be ideal for small toys made using 3D printing, as it can tolerate inaccuracies.
	Spur gears The most common type of gear, with straight teeth that are parallel to the gear axis. Spur gears are used to transmit power between parallel shafts and can have different ratios of teeth to adjust the speed and torque of the output shaft.
	Helical gears These gears have teeth that are angled to the gear axis, but also have a helical shape that allows for smoother and quieter operation. Helical gears are used to transmit power between parallel shafts and can have different ratios of teeth to adjust the speed and torque of the output shaft.
	Bevel gears These gears have teeth that are angled to the gear axis, allowing for the transmission of power between shafts that are at an angle to each other. Bevel gears can have different ratios of teeth and can be used to adjust the speed and torque of the output shaft.
	Worm gears These consist of a worm gear and a worm wheel, where the worm gear has a screw-like thread that meshes with the teeth on the worm wheel. Worm gears are used to transmit power between non-parallel and non-intersecting shafts and provide high reduction ratios.
	Planetary gears These gears consist of a central sun gear, several planet gears, and an outer ring gear. Planetary gears provide high reduction ratios and are commonly used in automotive and industrial applications.

If you want to learn how gears work and how to design them using Fusion 360, you can refer to following playlist:

- <https://www.youtube.com/playlist?list=PL4Yme23PQa9njUgvH2o2HP03-4RtZObzO>

Gear Ratios

	A is a driver gear and has 12 teeth (aka pinion), B is a driven gear and has 24 teeth. $\text{Gear Ratio} = \frac{T_{\text{driven}}}{T_{\text{driver}}} = \frac{T_B}{T_A} = \frac{24}{12} = 2:1$ It means there is a twice reduction in speed and a twice increase in torque. For example, A torque is 1 Nm and 200 rpm => B torque is 2 Nm and 100 rpm.
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	A is a driver gear and has 24 teeth, B is a driven gear and has 12 teeth. $\text{Gear Ratio} = \frac{T_{\text{driven}}}{T_{\text{driver}}} = \frac{T_B}{T_A} = \frac{12}{24} = 1:2$ It means there is a twice reduction in torque and a twice increase in speed. For example, A torque is 1 Nm and 200 rpm => B torque is 0.5 Nm and 400 rpm.
	This gear train transmits motion in the same direction through B idler gear. $\text{Gear Ratio} = \left(\frac{T_B}{T_A}\right) \left(\frac{T_C}{T_B}\right) = \left(\frac{12}{24}\right) \left(\frac{24}{12}\right) = 1:1$ Besides direction, torque and speed of A and C will be the same.
	A is a driver. Hence: $\text{Gear Ratio} = \left(\frac{T_B}{T_A}\right) \left(\frac{T_D}{T_C}\right) = \left(\frac{60}{8}\right) \left(\frac{52}{8}\right) = 48.75:1$ It means A has to turn 48.75 times to make full revolution of D which will increase D's torque 48.75 fold.

Important: Use of gears and gearboxes is **not compulsory**, but they can significantly amplify the carriage force of your vehicle. If you decide to design your own gearbox, take into account following precautions and hints:

- Measurements in your 3D model and 3D printed gears may vary. **Leave about 0.3 mm clearance** between any solid bodies that are supposed to interact. Not leaving enough clearance will result in parts not fitting together.
- Designing gears is a complex task taking into account several factors such as shape and size of teeth, pitch circle, pitch point, pressure angle, direction of shafts etc. Therefore for ease of design, the use of **spur gears** is recommended. They can be designed automatically using Fusion 360. Make your spur gears at least **5 mm thick** and **infill** at least **40% with 2 walls** around the edges, so they can sustain high load.
- The use of **worm gears** is recommended if you want to achieve a high reduction at once such as 20:1 or more. In this case, make your worm screw extremely durable for wear and lubricate contact points with silicone lube (e.g. WD-40 spray).
- Use your 100 gr limit for 3D printing wisely and don't waste it on other printing non-functional parts such as car body or wheels. You can use other materials for your car's body or wheels. Only use 3D printing for custom parts such as gears or gearbox.
- Print shafts and gears separately, so they can be replaced with other alternative gears. It will make your gearbox customizable and easily fixable.

Target Requirements of Vehicle

Your prototype should consist of a chassis, DC motor, 3 or 4 AA batteries (4.5-6V) and toggle switch to turn it on/off. It may also have other parts which are necessary for better stability, durability and performance of your prototype. The parts of your prototype can be either **3D printed** or **handcrafted**. Your prototype should operate on a bumpy inclined terrain. It also needs to have an exquisite appearance (you can style it your own way to make it look cool) and firm finish.

Below are the general technical requirements for your prototype to qualify for an experiment.

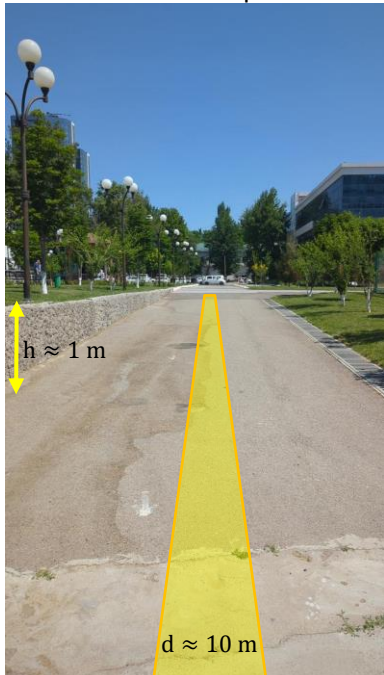
Parameter Name	Measurement
Max dimensions	Length: 40 cm Width: 20 cm, Height: 20 cm
Max input voltage	6 V
Min mass	50 grams
Battery capacity range	400 - 2850 mAh
Min speed	0.2 m/s
Operating ramp angle range	0° – 10°
Operating ramp length	0 – 100 m

Your prototype may have additional technical parameters depending on the design (e.g. number of wheels, radius of the wheel, number of gears and their technical characteristics, etc). You must provide these parameters in your presentation file along with more detailed final specifications of your own prototype.

Experimental Setup

In final experiment, we will measure the Output Power (P) of your prototype and rank it along with others' values in a descending. Your prototype should be fine-tuned for optimal P , because 50% of your project's mark will depend on your place in overall ranking. Check evaluation criteria section for more details.

Below are the steps of conducting the final experiment:

Sloped track at IUT's territory will be used for experiments  $h \approx 1 \text{ m}$ $d \approx 10 \text{ m}$	1. Check if prototype qualifies for the experiment and meets above technical requirements. 2. Measure the total mass (m) of the prototype on electronic scales. 3. Measure the total time spent (t) to reach the finish line on sloped track via stopwatch. 4. Calculate $P = \frac{mgh}{t}$ and save it 5. Calculate the efficiency of the prototype by dividing P to motor's maximized output power (0.14 W). It is done only for reference.
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Modelling Speed as the function of Mass

You can run your own experiments as described above. The main reason why you need to conduct your own experiments is to gather experimental data which you can use for optimizing the your prototype. For that you must find a polynomial function which can approximate the conversion of your vehicle's mass into your vehicle's speed. You can pick a polynomial function of maximum 3rd degree and not more. Hence it can be represented in terms of 4 coefficients a , b , c and d .

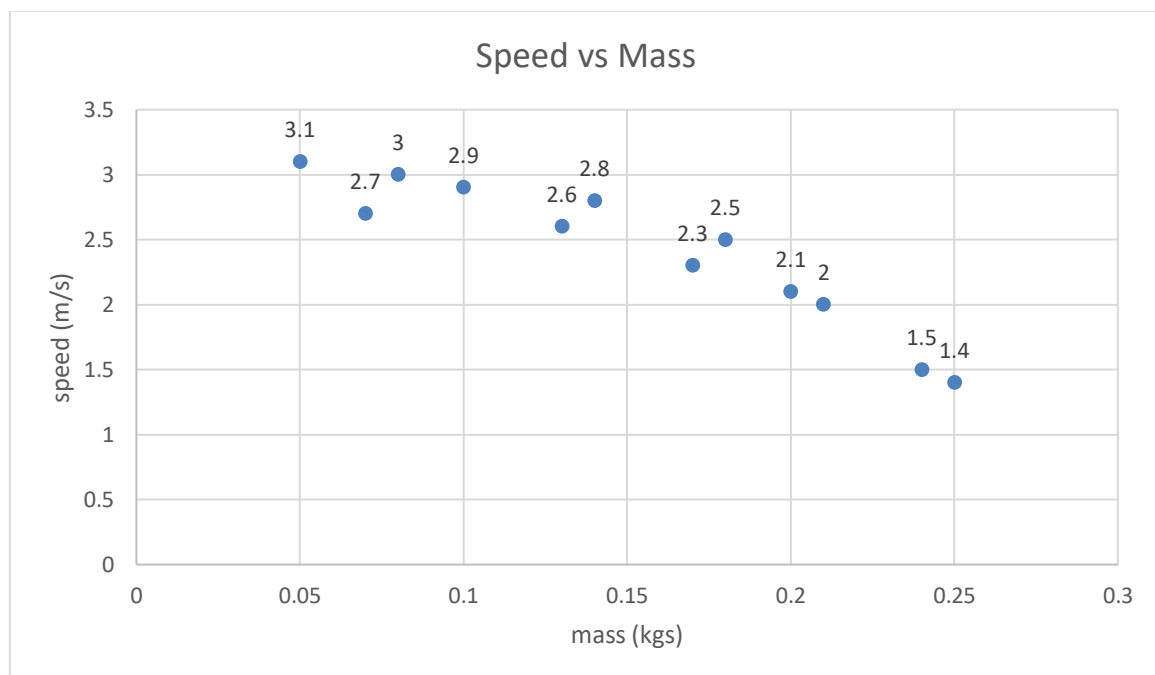
$$v_i = am_i^3 + bm_i^2 + cm_i + d$$

where v_i is the modelled speed and m_i is the mass for i th reading.

How can you find coefficients of your model? First, you must collect at least 10-15 readings (or more for better generalization) of speed for different masses. For example, you can add or remove load to your vehicle and measure the speed. Write down your readings in table form

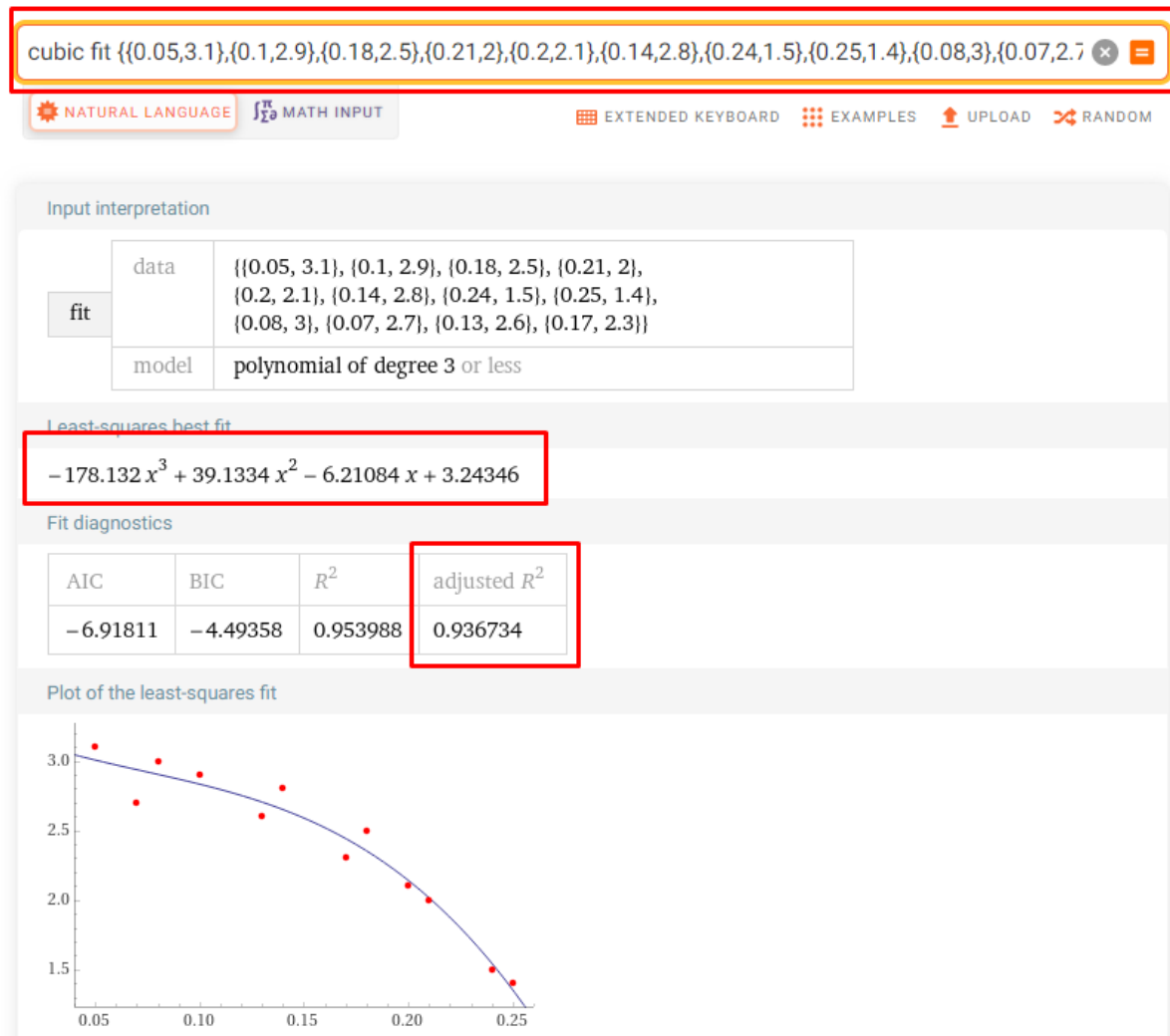
Reading #	Mass	Speed
0	0.05 kgs	3.1 m/s
1	0.10 kgs	2.9 m/s
2	0.18 kgs	2.5 m/s
3	0.21 kgs	2.0 m/s
4	...	...

Then you can do an exploratory analysis by plotting 2 parameters: v_i measured speed and m_i the mass. Below scatter plot is the sample plot you could create to visualize the magnitude of changes happening.



You can now randomly pick 80% of your dataset and use it to fit a polynomial curve. You can try first with a *cubic fit*, then proceed to a *quadratic fit* and finally to a *linear fit*. You can use [Wolfram Alpha](#) to accomplish your curve fitting procedures. Figure below shows an example of cubic fit where the coefficients of cubic fit are calculated. It also shows R^2 coefficient which measures how observed value varies from the predicted model output. R^2 coefficient ranges from 0 to 1 where $R^2 = 0$ means that observed value cannot be predicted with a given model, and $R^2 = 1$ means that observed value can be exactly predicted with a given model. However, if R^2 is too close to 1, it also signals that the model is overfitted to given instance of collected data and does not generalize the physical relationship between 2 parameters, namely: speed and mass. In order to avoid overfitting, you should

lower the degree of the polynomial fit. **Also, check that your coefficients are not too small (less than -10^4) or too big (greater than 10^4).** If they are too small or too big, your model is not reliable and cannot be used for predicting speed.



Once your polynomial function is found, you can use the remaining 20% of your dataset for testing the accuracy of your model. A good way of testing the accuracy of the model is using Residual Sum of Squares (RSS) as given below:

$$RSS = \sum_{i=1}^n (y_i - f(x_i))^2$$

The lower is RSS, the better your model generalizes the relationship between speed and mass. Never use the same dataset to train and test your model. Remember, while training your model (i.e. fitting the curve) you find a function which minimizes the RSS for a given dataset and while testing your model you check how this model would behave in wild.

Optimizing the Power

Once you establish a functional relationship between mass m and speed v , you can find such mass which would maximize the equation $P = \frac{mgh}{t} = mv \frac{gh}{d}$. Given that term $\frac{gh}{d}$ is constant, we need to maximize following equation:

$$\max_m m * v(m)$$

Taking the derivative of above statement would result in following equation whose solution would find optimal mass which maximize the power.

$$v(m) + m * v'(m) = 0$$

Note: You can do similar experiment and analysis for different gear ratios and speed. Gears modify torque and thus impact speed. Plot the resulting speed for given torque and model the results to find the optimum gear ratio which maximize the power.

Project Logistics

How to Change Your Team?

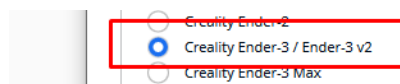
We will assume that teams will remain the same after completing the Project 2. However, if you still want to change your team or form a new team, you must email me about your change before deadline. I will make the change in team registration list on your behalf. Students who changed their teams but failed to contact me **will get 0** for this assignment.

Students who could not find a team **MUST** contact me by email **before above deadline**, so I can assign them to a team myself.

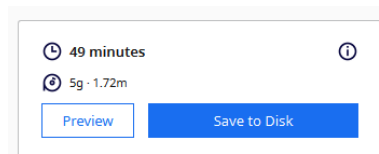
How to 3D Print Your Prototype?

You will print your 3D models yourself during your designated 3-hour timeslots when you can access IUT's 3D printers. There are five **Creality Ender 3** printers available in room A309 and the access to one of these printers will be given to your team starting from the next week. We will publish the schedule for each team in Telegram channel and post it on the door of A309 room. You should come with your team at a designated time and print your models.

Before coming to 3D printing room, you must prepare your 3D model for printing. This involves downloading and setting up Cura on your computer. Also you should add "**Creality Ender-3 / Ender-3 v2**" printer to your Cura. You can find more detailed video instructions in a QR code attached to printers as well.



Once you finish adding printer, you can drag-n-drop your STL models to Cura, rotate them to right orientation so it minimizes overhangs and maximizes adhesion and finally slice them according to your specific needs. After finishing the slicing, **check time** and **weight** at the bottom right of the screen. **Time** should not exceed **3 hours** and **weight** should not exceed **100 grams**.



If everything looks fine, you can save your sliced model to SD-Card available in your 3D printer and start the printing process. The video instructions for preheating and printing using **Creality Ender-3**

will be available in a QR code attached to the printer. After printing is finished, you should collect your model and turn off the printer.

The printing process will be overseen by me and my TA. If we find any misuse of 3D printers or negligence of above instructions, we will revoke your team's right to print or even suspend the 3D printing process for every teams. Then you may want to follow Plan B as given below.

Plan B:

If for some reason you were not able to print your 3D parts using IUT's facilities, you can contact private companies listed below for this service. They usually charge about 1200 soums per gram, but everything depends on the complexity of your parts.

- <https://3dbozor.uz>
- https://www.instagram.com/3d4p_tashkent
- <https://unit3d.uz/services/printing>
- <https://t.me/at3dprint>
- <https://t.me/iqshopuz>

Plan C:

If for some reason you were not able to print your 3D parts via third-party companies, you can handcraft them manually using wood, plastic, or any other suitable materials.

How to Test Your Prototype?

Once your working prototype is ready for testing, you can start doing your experiments in designated IUT's territory with sloped track. The purpose your prototype's tests is mostly the collection of data points for further modelling as described above. By trying out different configurations of your prototype (various gear ratios or weights), you may find the optimal balance between speed and torque which will maximize the power. The more data you gather, the better your model will generalize the relationship between speed and mass, and hence you are likely to the optimal configuration for your **FINAL EXPERIMENT**.

How to Complete Your Project?

You should organize group meetings every 2-3 days to assign tasks to members of the team and crosscheck on the progress of your project. Your meetings should be documented by filling out a **meeting minutes document** enclosed to this assignment. Below are the important steps in your project:

1. Review project specification document with your team carefully, and contact me by email if you have any questions.
2. Plan your project's major tasks using Gantt chart
3. Generate your concept sketches on paper using techniques such as SCAMPER, analogy, worst possible idea, brainstorming, mind mapping, morphology. Ensure that these concepts meet your product's target specifications. Apply decision matrix to select most promising ones and iterate at least 2-3 times.
4. Research literature on the subject matter, and identify technical or physical contradictions involved in realizing them. Use TRIZ for resolving these technical contradictions.
5. Develop high-fidelity 3D model of your final prototype using a CAD software. Design its assembly of parts in a CAD software. Make certain parts detachable for easier adaptation while testing your prototype in an experimental setup.
6. Prepare your parts for 3D printing. Convert them into STL format and estimate the amount of filament to be used for printing them. Follow IUT's guidelines for 3D printing.
7. Go to 3D printing room and print your prototype
8. Once your parts are ready, assemble your prototype and start testing your prototype.

9. While testing your prototype, collect as much data as possible. Save your data points in spreadsheet for later analysis.
10. Complete full exploratory analysis and modelling of your speed-mass function. Use 80% of your dataset for training and 20% for testing. Pick the best speed-mass function according to the least RSS.
11. Optimize power equation with your speed-mass function and find the optimal mass. Change your vehicle's mass accordingly and validate your analytical results with more experiments.
12. Reflect on your design process. Find strong and weak points of your work flow and proposed concept.
13. Prepare presentation slides as specified in this document.
14. Submit electronic version of your presentation file, meeting minutes and all supplementary materials to eClass before the deadline.
15. Print and bind your presentation file and meeting minutes along with relevant supplementary materials.
16. Bring your printed materials and prototype to **FINAL EXPERIMENT**.

How to Submit Your Project?

You should archive your presentation slides and supplementary materials into a single file named as your team name (e.g. Lions.zip). This file should be submitted **only ONCE** by the team leader to eClass **before deadline**.

You should print out your presentation slides and bind them to a folder. For each meeting you have conducted with your team members, you should also print a meeting minutes document signed by attendees. Anything relevant used or created during your meeting should be enclosed to your meeting minutes document. All your printed and signed meeting minutes and their relevant attachments must be bound to a folder and submitted as supplementary materials to your presentation slides. **All your printed documents should be given to me before your FINAL EXPERIMENT.**

How to Prepare Your Presentation Slides for Submission?

Oral presentation is not required in this project. However, you must prepare presentation slides which reflect your design process and empirical findings and submit them to eClass.

The structure of your presentation should be as shown below:

1. **Title page** contains team name, project title, team members' names and student IDs, section number.
2. **Introduction page(s)** provides your key research findings related to the subject matter. It may include theoretical background on relevant fields of physics.
3. **Concept generation page(s)** contains key highlights of your concept generation process and featured concept sketches you have initially proposed. You should highlight the use of concept generation techniques like SCAMPER, morphology, mind mapping, etc.
4. **Concept selection page(s)** contains methods of evaluating proposed concepts through decision matrices and iterations.
5. **TRIZ resolution page(s)** contains key technical or physical contradictions among your design metrics and the use of TRIZ guidelines in their resolution.
6. **High-fidelity 3D model page(s)** demonstrates 3D model of your proposed prototype. It should illustrate individual parts of your proposed prototype and the way how they assemble. Also you should enclose the technical drawings of your prototype.
7. **Experiments page(s)** contains the data you have collected through experiments in table form.
8. **Analysis page(s)** contains the scatter plot of your data and analysis of the relationship between speed and mass. It should show the definition your speed-mass function and the way how it was selected. Also your optimization of power based on speed-mass function should be shown.
9. **Validation page(s)** contains experimental data validating the result of your optimization.

10. **Conclusion page** discusses about challenges and weak parts of the proposed concept. It also provides reflection on the entire process of your team's work as a product development team.
11. **References page** provides links to relevant literature, 3D models, and all other online resources used in designing your product.

While preparing your presentation slides, you should avoid placing too much information into a single slide, placing irrelevant images from clip-art, using dark background or unreadable font size or colour, applying distractive animations like fancy slide transitions or effects. Ideally, font size should be 24 pts and slides should contain at most 4 to 5 bullet points per page.

How Your Project will be Assessed?

In general, projects will be evaluated along the following aspects that contribute to the overall quality of the accomplished work:

- Compliance with the product design process
- Feasibility of proposed product with current technology
- Evolution of concepts from quantity to quality
- Originality of the proposed product
- Quality of the presentation slides and supplementary materials
- Conformance to the technical requirements of physical prototype
- Accuracy of presented physical prototype

The marks for this project will be distributed according to following scheme:

Assessment Component	Mark (points)
Introduction	5
Concept generation and selection	10
TRIZ resolutions	5
High-fidelity 3D model	5
Experiments, exploratory analysis and conversion functions	10
Conclusion and self-evaluation	5
Quality of submitted materials (presentation slides / supplementary materials)	10
Measuring output power of the physical prototype	50
After conducting the final experiment for all teams from all sections, the POWER of each team's prototype will be sorted in a descending order and points will be allocated according to team's respective rank in the list. The top team will get 50 points, next team 49 points, and next 48 points, and so on.	
Total	100

Penalties

Teams are subject to a penalty or disciplinary action for reasons of academic dishonesty and negligence. Academic dishonesty can be any attempt by a student to submit 1) work completed by another person outside the team without proper citation, 2) or give improper aid to another student outside the team in the completion of an assignment. Academic negligence can be disregarding the key requirements of the project including the timely submission of the project's deliverables. Following penalties will be exercised if any of the below cases happen:

- **Late submission.** If you missed the deadline, you must send your project deliverables over email to s.abdullaev@inha.uz within the next day after the deadline. You will be penalized by **20 points** for late submission. **Important notice:** you are not required to send your project deliverables by email if you have already submitted them to eClass before the deadline. If you submit your project deliverables to 2 places: eClass and email, you will get a penalty of **5 points** for confusing the assessment.
- **Making noise while others presenting.** If any member of your team is spotted for making noise while others presenting, the entire team will be penalized by **20 points**. If this repeats again, the team will be requested to leave the classroom and their mark will be 0.
- **Not informing about team change.** If you change a team or unable to find a team, you should inform me by email about this fact before team registration deadline. I will make changes in my teams list or assign you to a different team. If you fail to inform this fact or inform it after the deadline, you will be penalized by **10 points**.

Any other academic misconducts including plagiarism, cheating, or breaching internal rules and regulations of IUT will be brought to the attention of Academic Committee for Penalty and Awards.

Disclaimer:

- I reserve the right to grade each member of the team separately if I have a suspicion on his/her contribution to the project. I will review meeting minutes submitted with the project, and if I find a member who has not contributed to the project, I will call this member for a short viva regarding his/her contribution. If member fails to show up, he/she will receive 0 for the project.
- I reserve the right to call the whole team for a short viva if I have suspicion about the quality of work submitted. If team fails to explain how web application has been implemented, each member of the team receive 0 for the project.